Pipe Threading Machine



Category: **Plumbing & Pipe Fitting** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Powered pipe threading machine

What it is used for

A powered pipe threading machine cuts external threads on compatible pipes so that fittings can be securely connected. Correct die selection, stable pipe clamping, and controlled operation help produce accurate threads while reducing injury and equipment damage.

Main parts

- Motor drives the threading head and rotating pipe or die mechanism.
- Chuck or clamping jaws hold the pipe securely.
- Threading head holds the dies that cut the threads.
- Die set determines the thread size and type.
- Pipe support or stand supports longer pipe sections.
- Cutting oil system lubricates and cools the cutting area where fitted.
- Foot switch or control switch operates the machine on applicable models.
- Chip tray collects metal chips and excess cutting oil.

Before you start

1. Read the manual and confirm the machine supports the pipe material, diameter, and thread type.
2. Inspect the power cable, chuck, dies, guards, foot switch, and oil system for damage.
3. Select a compatible, sharp die set of the correct size.
4. Check that the pipe is straight enough, free from serious damage, and suitable for threading.
5. Secure the machine on a stable surface and support long pipes with suitable stands.
6. Wear safety goggles and close-fitting work clothing; remove jewellery and secure long hair.

How to use

1. Switch off and isolate power before fitting dies or adjusting the chuck.
2. Install the correct die set and set the threading head according to the manual.
3. Insert the pipe into the chuck and clamp it securely with sufficient length supported.
4. Position the pipe support so the pipe remains level and stable.
5. Check that guards are in place and the cutting oil supply is ready where fitted.
6. Start the machine using the designated control and apply cutting oil as instructed.
7. Guide the threading head into the pipe end without forcing the dies.
8. Allow the machine to cut the thread at the recommended feed and speed.
9. Stop the machine and wait for all rotating parts to stop before inspecting the thread.

10. Release the pipe only after rotation has stopped, remove chips safely, and check thread fit with the specified gauge or fitting.

⚠ Safety

- Never wear gloves, loose clothing, or jewellery near rotating pipe or chuck components.
- Keep hands away from dies, chuck jaws, and rotating pipe during operation.
- Do not hold long pipe sections by hand alone; use suitable pipe supports.
- Switch off and isolate power before clearing chips, changing dies, or servicing the machine.
- Never operate the machine with missing guards, damaged switches, or loose clamping jaws.
- Use a brush or approved tool to remove metal chips; do not use bare hands.

Care & storage

- Clean metal chips and cutting oil from the machine after use.
- Inspect dies for wear, chipped cutting edges, and correct fit.
- Check chuck jaws, pipe supports, guards, and oil delivery regularly.
- Store dies clean and dry and maintain the machine according to the manual.

Common mistakes to avoid

- Using the wrong die size or thread type.
- Clamping the pipe loosely or failing to support its free end.
- Touching or removing chips while the pipe or chuck is rotating.
- Forcing the threading head instead of using the recommended feed.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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